

Serverless Student Performance Analytics System

Solution Design Document (SDD)

Project Information

Item	Details
Project Name	Serverless Student Performance Analytics System
Prepared By	Shreyash Singh
Platform	Amazon Web Services (AWS)
Programming Language	Python
Database	Amazon DynamoDB
Storage Service	Amazon S3
Compute Service	AWS Lambda

1. Project Overview

The Serverless Student Performance Analytics System is designed to automate the collection, processing, storage, and analysis of student academic performance data. The solution uses AWS serverless technologies to process uploaded datasets without requiring dedicated servers.

Faculty members upload student datasets to an Amazon S3 bucket. Once a file is uploaded, an event notification automatically triggers an AWS Lambda function. The Lambda function validates the dataset, calculates student performance metrics, assigns performance categories, and stores the processed records in Amazon DynamoDB.

The system enables educational institutions to monitor student performance, identify academically at-risk students, and generate analytical insights efficiently.

The system supports:

- Automated data ingestion
- CRUD operations
- Performance analysis
- Student risk detection
- Leaderboard generation
- Data export functionality
- Advanced querying using DynamoDB Global Secondary Indexes (GSI)

2. Problem Statement

Educational institutions often maintain student performance records in spreadsheets or local databases. This approach introduces several operational challenges:

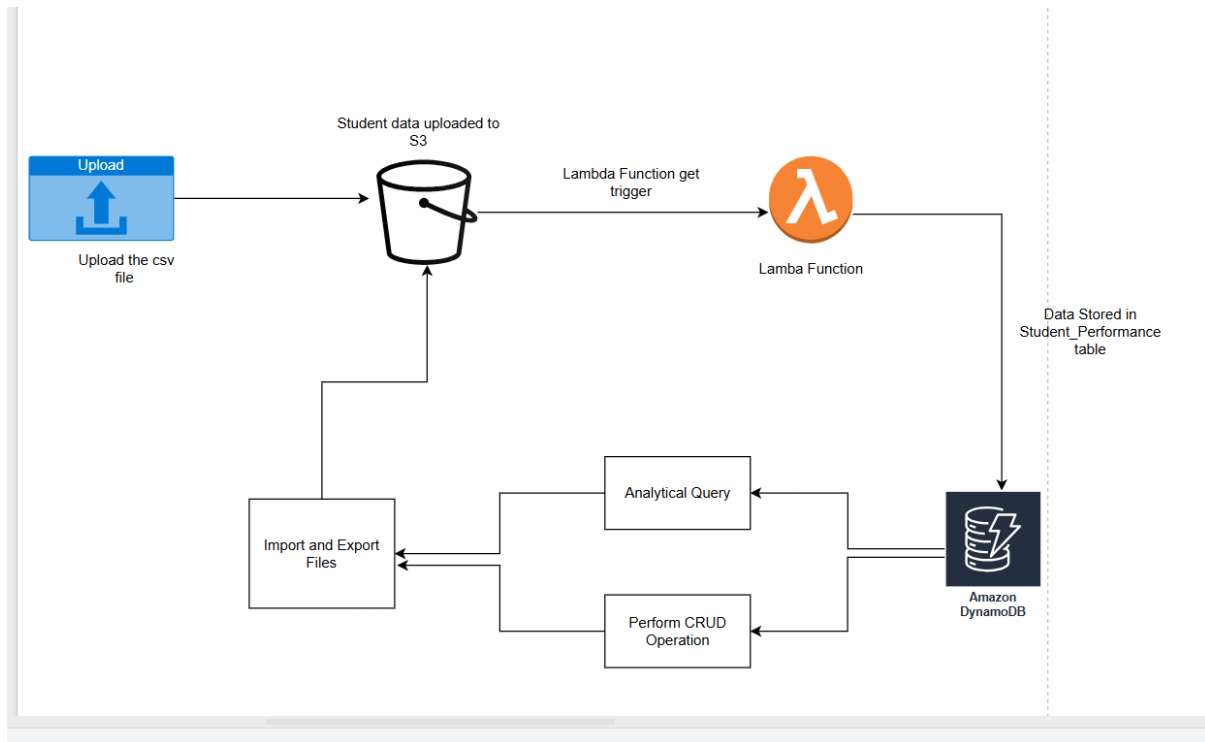
- Manual data entry and validation
- Delayed performance analysis
- Lack of centralized data storage
- Difficulty identifying low-performing students
- Limited reporting capabilities
- Increased maintenance effort

Objectives

The system aims to:

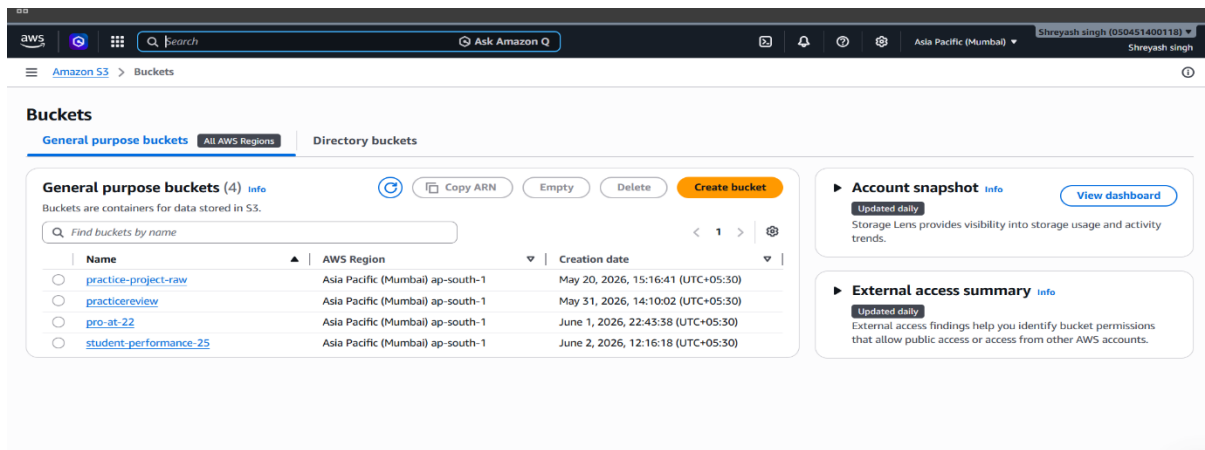
- ✓ Automate the ingestion and processing of student performance datasets
- ✓ Store and manage student records efficiently using Amazon DynamoDB
- ✓ Analyze academic performance based on attendance, study hours, and class participation
- ✓ Categorize students according to their overall performance scores
- ✓ Identify students who may require academic intervention
- ✓ Generate performance insights and leaderboard reports
- ✓ Enable efficient retrieval and management of student data through CRUD operations
- ✓ Provide a scalable and cost-effective solution using AWS serverless services
- ✓ Reduce manual effort and improve the speed of academic data analysis

4.High Level Architecture

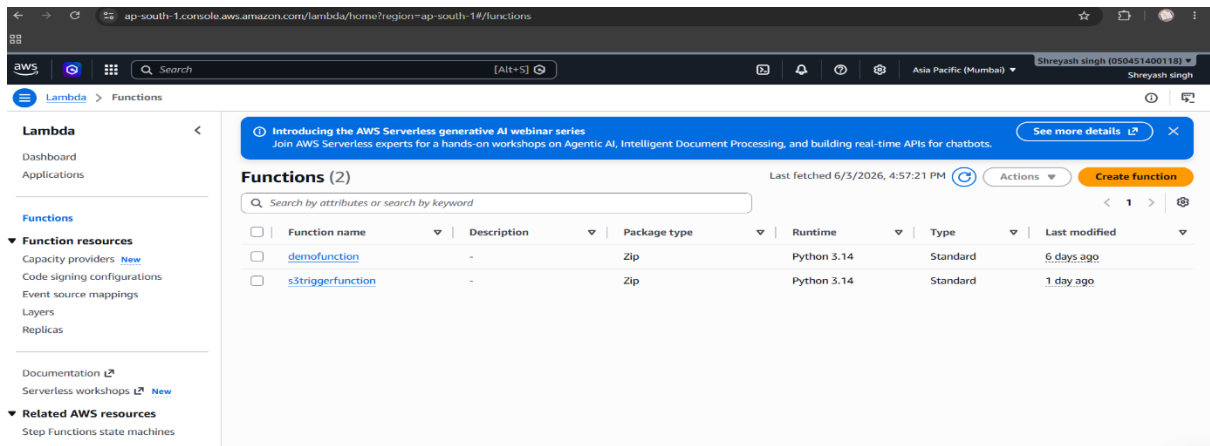


5. AWS Services Used

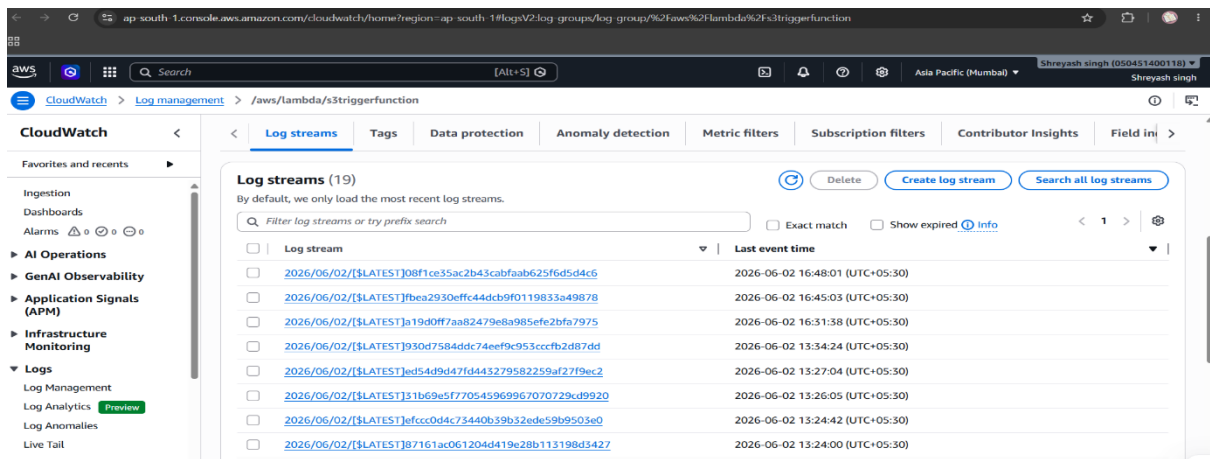
AWS S3 Storage



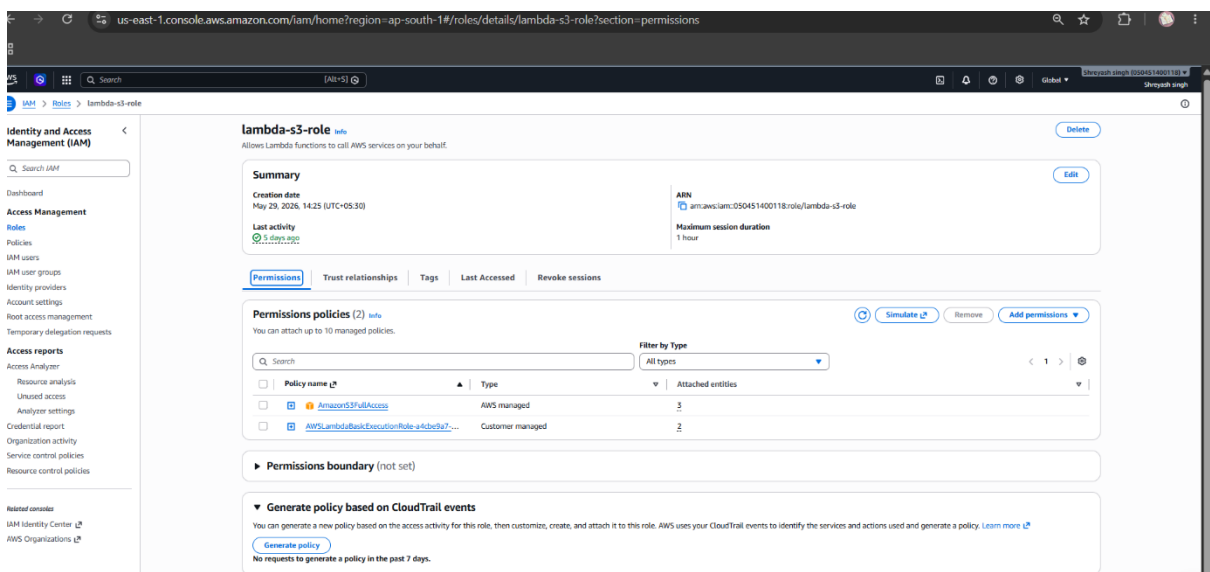
Lambda Functions



CloudWatch



IAM Role



DynamoDB Table

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↺

ap-south-1.console.aws.amazon.com/dynamodbv2/home?region=ap-south-1#tables

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Asia Pacific (Mumbai)

Shreyash singh (050451400118)

Shreyash singh

DynamoDB

> Tables

DynamoDB

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Dashboard

Tables

Explore items

PartiQL editor

Backups

Exports to S3

Imports from S3

Integrations

Reserved capacity

Settings

DAX

Clusters

Subnet groups

Parameter groups

Events

🟢 The request to delete the "Practice1, Students2" tables has been submitted successfully.

×

Tables (4) info

Last updated June 3, 2026, 16:58 (UTC+5:30)

🔄

Actions

Delete

Create table

▼

🔍 Find tables

Filter by tag Any tag key ▼

Filter by tag value Any tag value ▼

< 1 >

⚙️

☐	Name ▲	Status ▼	Partition key ▼	Sort key ▼	Indexes ▼	Replication Regions ▼	Deletion protection ▼	Favorite ▼	Read
☐	Employee	🟢 Active	eid (S)	-	0	0	🔒 Off	☆	On-d
☐	student_performance	🟢 Active	student_id (S)	-	1	0	🔒 Off	☆	On-d
☐	Students	🟢 Active	Sid (S)	-	0	0	🔒 Off	☆	On-d
☐	table1	🟢 Active	id (S)	-	0	0	🔒 Off	☆	On-d

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